AMMAR FITWALLA

Senior AI/ML Engineer

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Senior AI/ML Engineer with **7 years** of end-to-end experience — from raw data pipelines to production-grade APIs and cloud deployment. Built and shipped ML systems across fintech, healthcare, adtech and telecom domains at both **enterprise scale** (Ericsson, WPP) and startup speed. Independently developed and launched live **AI** products including an **LLM** forensic platform and **Agentic AI** campaign intelligence system — because I build outside work hours too. Strong across the full stack: **Python, TensorFlow, PyTorch, LangChain, AWS, GCP, FastAPI, Docker**.

FEATURED PROJECTS

Justitia Lens — Multimodal AI Forensic Analysis Platform | Google Hackathon Project

- **Top Submission (Out of 4,499+):** Solo-built and deployed a live **multimodal AI** platform on GCP that detects inconsistencies between police reports and visual evidence (photos/bodycam footage).
- **Cross-Modal Reasoning:** Implemented complex prompt orchestration using **Gemini 3.0 Flash** to align timestamps and verify claims across long-context text, images, and video.
- **Structured Outputs:** Designed schema-guided LLM generation to produce verifiable, hallucination-free forensic audit reports.
- **Full-Stack Architecture:** Engineered the complete **Next.js** frontend, **FastAPI** backend, and **PostgreSQL** database to enable automated evidence auditing workflows.

JobAwn — AI-Powered Job Search & Referral Platform | Live Deployment (jobawn.com)

- **Resilient Resilient Data Pipeline:** Built a concurrent data scraping pipeline with automatic rate limiting and circuit breakers, scaling a live SQLite database to 50K+ jobs.
- **Fault-Tolerant AI Matching:** Built a multithreaded resume–JD matching engine with LLM provider fallback and intelligent caching, reducing redundant inference for repeated evaluations.
- **Privacy & Multi-Tenant Architecture:** Built a PII-redaction pipeline validated by 27-unit tests, sanitizing sensitive data before external LLM transmission; designed isolated per-user sessions, resume parsing, and scoring state to support scalable multi-tenant execution.
- **Production Reliability & Ops:** Deployed the FastAPI/vanilla-JS stack on Oracle Cloud via Docker, integrating monitoring and automated SMTP backups; **mitigated LLM quota exhaustion and SQLite write contention** through client-side rate limiting, targeted retries, coordinated writes, and zero-downtime schema migrations.

WORK EXPERIENCE

Senior Associate — Projects | Cognizant — Client: Google

April 2026 - Present

Tech Lead, AI-Enabled Operations Platform

- **Platform Platform & Team Leadership:** Led a 3-engineer backend team in deploying an AI-powered analytics platform, scaling operations across global TSRs and vendor partners.
- **LLM Latency & Orchestration:** Enhanced real-time WebSocket streaming for chunk-by-chunk token delivery, achieving sub-second (< 500 ms) perceived user latency. Implemented asynchronous parallel tool execution (asyncio.gather) across BigQuery and analytics tools to sustain non-blocking multi-agent workflows.
- **Agentic Tool Calling Architecture:** Architected a dynamic function-calling pipeline (using **Google ADK and Gemini 2.5 Pro**) where the **LLM** evaluates user intent to trigger specialized SQL execution tools, seamlessly feeding the structured database outputs back to the model for natural language summarization.
- **Enterprise AI Resiliency:** Hardened **Vertex AI** integrations against 429/500/503 rate limits and quota exhausts by implementing an 8-attempt exponential backoff with random jitter (tenacity), preventing thundering herd failures during peak enterprise load.
- **Security & Guardrails:** Engineered strict anti-jailbreak guardrails via system prompts to securely orchestrate tool calling and bound LLM behavior. Maintained ultra-low overhead pipeline security with real-time DLP (Data Loss Prevention) inspection operating at ~2–50 ms latency.
- **Continuous Risk Detection & Precision Tuning:** Refined an automated risk-detection service that continuously analyzes BigQuery interaction data using structured LLM reasoning. **Improved detection precision by reducing false-positive "smoky" case flags from ~35% to under 10%**, eliminating alert fatigue and effectively replacing manual reviews.

Senior Manager — Data & Analytics | WPP Media

September 2025 – March 2026

World's largest advertising group — India/APAC media analytics team

- Led full data science lifecycle optimizing **XGBoost**-based campaign forecasting model for media client campaigns — improving prediction accuracy and reliability for multi-day ad performance forecasting.
- Built automation suite for Ops team, **saving 240+ hours annually** through automated data processing, data migration, validation and standardization. Developed multi-report automation suite projected to save an additional **5 days** monthly.

- Built **Agentic AI** chatbot using **LangGraph** multi-agent pipeline (Planner → Executor → Insights nodes) for natural language querying of campaign data — enabling ops team to query millions of rows of media campaign data in plain English without SQL knowledge, running fully local via Ollama with zero data leaving the machine. Planner node parses user intent, Executor runs SQL **tool-calling** against campaign database, Insights node generates business narrative from raw results. Integrated into production PyQt5 desktop application used by WPP Media ops team.

System Engineer (AI/ML) | **Tata Consultancy Services — Client: Ericsson** *March 2022 – August 2025*

3.5 years | Enterprise-scale telecom client engagement

- Built **time-series forecasting models (ARIMA, Prophet)** with automated daily prediction pipelines via **Jenkins** — serving forecast outputs to **50,000+** Ericsson Business Users globally for telecom business trend analysis and operational resource planning.
- Built and maintained ML data pipelines and automated cron jobs on Jenkins infrastructure — reducing manual intervention by **80%** and ensuring reliable, timely data delivery to downstream production models.
- Exposed ML prediction services via **Python Flask** REST APIs — consumed by internal business and engineering teams for operational decision making.
- **TCS Star of the Month Award** — September 2022.

Data Scientist | **KernelSphere Technologies LLC** *January 2021 – February 2022*

Early-stage funded AI/ML startup

- Engineered a **production-grade** Forex trend prediction system — trained **LSTM neural networks** on **30GB+** of historical data alongside **Prophet** and **XGBoost** models, deployed on **AWS SageMaker** as real-time APIs, and served to paying client making active trading decisions via a **Django** web application on **EC2**.
- Independently built **NLP models** to extract structured financial data from unstructured PDFs, achieving **80% extraction accuracy**.

Machine Learning Engineer | **KernelSphere Technologies LLC** *June 2019 – January 2021*

Early-stage funded AI/ML startup

- Built **Deep Learning** models for healthcare disease classification — achieving **90%** accuracy and significantly reducing preliminary diagnosis time for clinical workflows.
- Applied **CNN**, **R-CNN**, and **YOLOv3** algorithms to detect architectural objects in high-resolution civil engineering drawings (~3000×3000 pixels) with **~80%** accuracy.
- Designed **custom algorithms** to extract and map blueprint data into a **GraphQL database** — replacing manual data entry and enabling structured, fast retrieval of architectural information at enterprise scale.

SKILLS

AI / ML	Python, TensorFlow, PyTorch, Scikit-Learn, XGBoost, LSTM, CNN, R-CNN, YOLOv3, ARIMA
LLMs & GenAI	LLMs, RAG, LangChain, LangGraph, Vector Databases, Prompt Engineering, Multimodal AI
Web & APIs	Django, Flask, FastAPI, React.js, Next.js, REST API Development
Cloud & DevOps	AWS (S3, SageMaker, EC2), GCP (BigQuery, Cloud SQL, App Engine, Cloud Scheduler. Cloud Run Functions), Docker, Kubernetes, Jenkins, CI/CD
Databases & Tools	PostgreSQL / PostGIS, GraphQL, Git, Agile/Scrum

EDUCATION

Master of Engineering, Information Technology *2021*

University of Mumbai

Bachelor of Engineering, Computer Engineering *2017*

University of Mumbai

ACTIVITIES & RECOGNITION

- **Published Research**: Authored a paper in the International Journal of Engineering Research & Technology (IJERT).
- **Guest Lecturer**: Delivered talks on AI and IoT at Vidyalankar Institute of Technology, Mumbai.
- **Google Gemini API Developer Competition** — Solo built and deployed **Justitia Lens** among 4,499+ submissions.
- **TCS Star of the Month** — September 2022.
- **Open-Source Contributor:** Diagnosed an issue and submitted a patch for Pydantic serialization handling within the **google/adk-python** repository.